

Testing Oxygen Generation for Medical Application
Technical Report

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Executive Summary

This work set out to build a clear, defensible picture of how the prototype 13X bed behaves under adsorption and to translate those physics into design choices for a small medical oxygen concentrator. Using a 1-D model with Ergun momentum, axial dispersion, linear driving force kinetics and a single-temperature energy balance, two complementary tests were run. First, an adsorption-only purity snapshot at the product end returned 72.2 percent oxygen at 18 seconds.

Second, a breakthrough run from a regenerated gas phase gave a 10 percent nitrogen breakthrough at 6.87 seconds and a 50 percent breakthrough at 27.24 seconds. Temperature rose from 298 to about 306 along the bed as the front formed and moved, while pressure loss remained small at roughly 0.01 bar across 0.55 metres. Read together, these results show a bed that is hydraulically easy to drive, thermally active but not unstable, and limited by early nitrogen slip unless the produce window is tightly selected.

Against the clinical bar commonly used in standards and procurement, the snapshot purity under-achieves and the useful production window sits before the 10 percent breakthrough, not after it. The remedy is not a single tweak but a move to proper cycle modelling and measurement: flow-weighted purity, recovery and productivity over a defined produce cut at cyclic steady state. Once that is in place, standard levers are available to reach medical-grade output without radical hardware change: timing and purge optimisation, pressure equalisation between beds, modest shifts in particle size, superficial velocity and bed length, and, critically, temperature-dependent equilibria so heat release and working capacity are linked in the model in the same way they are linked in the device.

1.0 Introduction

1.1 Background and Motivation

Oxygen generators serve as an essential function in a multitude of healthcare environments, providing a steady and controlled supply of medical-grade oxygen to the patient. These devices remain vital for oxygen therapy, particularly for individuals that are plagued by respiratory problems. Asthma and pneumonia are just a few of the conditions that come to mind when delving into respiratory illnesses. Some lesser-known illnesses include (and are not limited to) chronic obstructive pulmonary disease and acute respiratory distress syndrome (Melloni et al., 2018). According to the World Health Organisation, medicinal oxygen should contain a concentration percentage of 82% and above.

1.2 Project Aims and Objectives

The project's original objective was to successfully complete the assembly of a solar- powered oxygen generator prototype that could be used within the healthcare industry whilst also running tests- using MATLAB- to investigate its efficacy. The oxygen generator prototype should have its results broken down and analyzed to allow for a discussion on potential modifications that could be made onto the entire system. This project was inherited from a fourth-year group that manufactured the initial prototype and –subsequently- from a student in the previous year. The group that originally formed the prototype were the individuals that were responsible for the integration of the pressure swing absorption (PSA) technique that was vital in the nitrogen removal process from air. Due to certain limitations, the prototype remained unable to incorporate solar power. After changes in circumstances, the aim now was to model a 1D oxygen generator and to also curate a piece of code that can allow for the digital model to be tested against a variety of objectives.

1.3. Project Justification

Oxygen generators remain a vital aspect in less developed areas for individuals that are dealing with severe respiratory issues. They became quite prominent during the COVID pandemic where there was a clear and dramatic increase in the demand for supplemental oxygen therapy. Patients recovering from COVID required large amounts of oxygen due to the damage that the virus causes in the body's respiratory system. Research indicates that roughly 15% of the hospitalized COVID patients required long-term oxygen therapy upon being deemed medically

fit to leave (Doğru et al., 2024). The project places an emphasis on attempting to develop more of a low cost solution to the increasing need for medical grade oxygen. The financial implications of incorporating oxygen generators into many care facilities can be quite significant, with prices varying widely based on their capacity and/or brand. Typical price ranges can be quoted to approximately range from £1500 to £4446.09 (Kumar et al., 2021). These prices make it quite difficult to have oxygen generators implemented in low income countries. Thus further highlighting the importance of offering a more sustainable and cheap solution through the utilization of cheaper materials and solar power.

1.4 Duration of Project

The project began around the early stages of October 2024 and has lasted until the 26th of August 2025(due to continuous circumstances throughout the year, the project had to be extended. Weekly meetings were also scheduled with my supervisor to allow for easier communication and project updates whilst also allowing for better project planning. Term one consisted mostly of understanding what needed to be done whilst also conducting research on oxygen generators and the workings of the current oxygen prototype. The start of term 2 was mostly used for manufacturing new zeolite holders, taking the measurements of the current prototype to have new wooden panels installed, fixing the compressor in a spot that could accurately allow it to read the amount of oxygen compressed.

2.0 Literature Review

2.1 Reactor Composition

The reactor in the context of this project refers to the individual adsorption canisters/beds that are filled with the zeolite substance. Lorenzo's oxygen generator relied on the use of pressure swing adsorption which refers to a method of separating gas that is purely dependent on the pressure of the total mixture as well as the specific affinity that each component has with the absorbent.

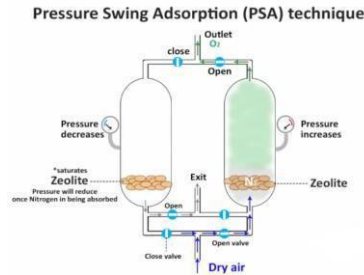


Figure 1

The vessels within the reactor system act as cyclic reactors that are able to alternate between the adsorption of nitrogen from compressed air and regeneration made possible via depressurization and purge. The two beds are operated in an antiphase system that utilises solenoid valves to effectively achieve the continuous supply of oxygen. An accurate non-physical system would require that the physical canisters are modelled precisely and potentially numerically as 1D packed bed reactors.

2.2 Modelling Equations and the Cycle in Literature

The dominant modelling approach for a PSA based oxygen generator was observed to be a one-dimensional axially dispersed plug flow framework; for the framework to function, assumptions were made. Radial gradients were assumed to be negligible and ultimately ignored; uniform bed properties were adopted into the model. Moreover, the gas mixture present was simplified to N_2/O_2 with nitrogen being the element strongly adsorbed and oxygen weakly adsorbed. Santos and a few other researchers developed a dynamic model for small medical pressure swing adsorption units through the use of various equalization configurations.

Adsorbents such as OxySiv 5, OxySiv 7 and Sylobead MS S 624 were compared to each other during the tests. The simulations were validated against the experimental data which generally portrayed good alignment with the performance metrics. Some of these metrics included product recovery and power consumption(Santos et al., 2006).

Typically, the modelling equations utilized in specific 1D models were found to be: axial dispersion, convective transport and adsorption sink terms based on a linear driving force approximation as well as Langmuir isotherms. Many studies assumed a negligible drop in pressure and isothermal conditions to aid in simplifying their models and to allow for moderate

cycle times(Cruz et al.,2007). The cycle steps involved included: adsorption, depressurisation, purging and repressurisation. These steps can be represented using boundary conditions that rely on time at the entrance and the exit of the bed. Furthermore, these steps originate from the Skarstrom cycle which is the foundational cycle idea that is used to execute pressure swing adsorption. Most modern generators fundamentally use this technique (with some having designs that improve performance metrics such as bed size factor and oxygen recovery) (Arora & Hasan, 2021).

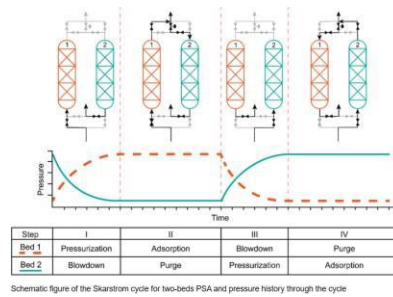


Figure 2

Interestingly, more recent reviews investigating the use of pressure swing adsorption modelling approaches highlight the importance of using simplified first-principle models(1-D and plug flow) during the early design and feasibility stages of oxygen generation. When used cooperatively with simulation-based frameworks, adsorbent productivity, recovery and compactness can all be enhanced (Carty & Bilodeau, 2023).

2.3 Axial dispersion

In adsorption-based reactors used when building oxygen generators, plug flow is almost always assumed for the simplification of the system. Under this assumption, gas molecules present in the reactors all travel at the same velocity along its axis, leading to a sharply defined separation front between the nitrogen-rich and oxygen-rich regions. When investigated, axial dispersion is found to occur due to the existence of two primary mechanisms.

Velocity gradients are found as gas moves throughout the pore spaces of the bed; molecules at the centre of the pores move quicker than those stunted by existing frictional forces near the surface of particles. Secondly, molecular diffusion occurs in parallel, leading to concentration gradients levelling out along the axis of the reactor. Both effects in tandem cause a phenomenon known as longitudinal mixing and lead to a broadening of the concentration profile within the bed. This occurrence can be referred to as axial dispersion (Ruthven, 1984).

2.4 Air Concentration Profiles

Air concentration profiles outline how the gas-phase composition(referring to the mole fractions of nitrogen and oxygen) and the corresponding adsorbed loadings change whilst moving along the axial length of a full zeolite bed as a function of the time it takes for a cycle to be completed. In one-dimensional zeolite bed models, these profiles reveal the movement of what's called the 'mass-transfer zone' which can be described as a finite region separating the 'fresh' and 'used' adsorbent. The adsorbent's properties are governed by principles and teachings mentioned previously such as axial dispersion.

In medical oxygen concentrators that utilise nitrogen selective zeolites, the adsorption step develops a nitrogen-rich mass transfer zone that propagates towards the product end. During this stage, the gas phase oxygen mole fraction rises as the nitrogen mole is specifically removed. While referring back to the Skarstrom cycle, the depressurisation and purge stages allow for full reversal of the mass transfer zone(front) and a change in the spatial gradients in the opposite direction. Heat effects are further compounded onto each other due to the nitrogen mole adsorption being an exothermic reaction and the desorption being an endothermic reaction (Arora & Hasan, 2021; Vemula et al., 2021).

Many 1D models incorporate heat effects and realistic dispersion to further increase the accuracy of the model. Parameters such as productivity and evolution of profile can also be monitored more efficiently. This was made apparent through measuring pressure and composition histories in pressure swing adsorption concentrators; sharper axial oxygen and nitrogen fronts at the end of pressurisation and adsorption processes were linked with greater product purity (Vemula et al., 2021; Sun et al., 2023).

To summarise, air concentration profiles within zeolite beds in PSA oxygen generation are moving axial fronts whose efficacy and position are dependent on factors such as kinetics, dispersion, particle size, porosity and step-timing. Most of these factors should be considered when building a credible and reliable simulation that accurately reproduces a mass transfer zone and the resulting oxygen purity at a steady and cyclic state (Ruthven et al., 1994; Sun et al., 2023).

3.0 Mechanical Setup

3.1 Prototype Composition

The physical prototype was studied to further improve the understanding of the project as a whole. Whilst being more of a modelling heavy project towards the end, it was important that the original design was observed to see just how oxygen generation could be achieved in a real world environment.

The mechanical design of the prototype was based on a compressor capable of producing compressed air up to 10 bar. In a model, this becomes the inlet condition that dictates the available flow rate and pressure for the system. Downstream of the compressor, two zeolite 13X canisters were arranged in parallel. These acted as adsorption beds, selectively trapping nitrogen while allowing oxygen to pass through. The presence of two beds allowed for alternating operation so that one vessel could generate oxygen while the other was regenerated. This arrangement, which was pressure-tested to around 9 bar, is fundamental to understanding how oxygen output can remain continuous across a cycle.

Airflow between these beds was controlled by a combination of in-line and 3-way solenoid valves, which switched open and closed according to a programmed sequence. In practice, these valves created a six-stage PSA cycle, with each stage lasting between 5 and 7 seconds. From a modelling perspective, this cycle provides a framework to define how flows alternate, how pressure rises and falls in each vessel, and ultimately how oxygen concentration at the outlet evolves over time.

The electrical system consisted of a 12 V, 50 Ah lithium iron phosphate battery, which supplied power to both the compressor and valves. An Arduino microcontroller was used to coordinate valve timing and compressor switching, assisted by relays and MOSFET boards. Although these physical components are not recreated here, their specifications are useful for estimating power demand and testing how different control strategies influence the performance of the cycle.

Solar charging was proposed in the earlier design but never fully implemented, meaning the prototype operated mainly from battery storage. This provides a simpler basis for simulation, focusing on the PSA cycle itself rather than broader renewable integration.

The ES410 group, who were responsible for the initial construction of the prototype, determined through their testing that both leakage and pressure integrity assessments had already been

completed to a satisfactory standard. Their evaluation demonstrated that the canisters were capable of withstanding internal pressures of up to 9 bar, with the onset of minor leakage only observed once pressures exceeded approximately 8.2 bar, specifically at the cylinder bolts and pipe connectors. Furthermore, when the canisters were pressurised to 8.2 bar and maintained under these conditions for a period of two hours, no reduction in pressure was detected, thereby confirming the absence of significant leakage pathways and validating the structural integrity of the system under the tested operating conditions.

3.2 Compressor

An important design improvement concerns the mounting of the compressor. In its earlier arrangement, the compressor was not firmly fixed to the lower base, leaving it susceptible to unwanted vibration and instability during operation. Various design ideas for securing the compressor were considered such as manufacturing a component akin to a “seatbelt” to hold the compressor in place. Eventually, a decision was made to keep the process simple and drill holes in the wooden base for the compressor to be bolted to.

Securing it to the base provided greater stability, which could potentially result in more reliable pressure readings and improved overall consistency of the system.

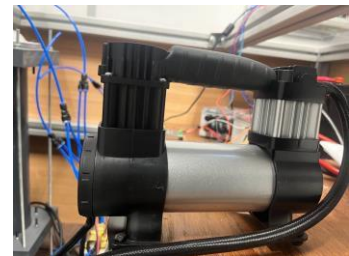


Figure 3

3.3 Filters

Zeolite filters were also constructed. The filters within the prototype were designed as circular disks manufactured from 3D-printed plastic, each incorporating a thickened rim to provide structural strength under operating pressures.

With a radius of 9.6 cm and a thickness of 0.4 cm, the disks were dimensioned to sit securely within the assembly while maintaining durability. To ensure adequate airflow, the interior of each disk was patterned with regularly spaced 2 mm square perforations, which not only supported consistent air passage but also helped distribute flow evenly across the surface. The filters were fixed in position between the flange and dome, providing both mechanical stability and functional reliability within the system.

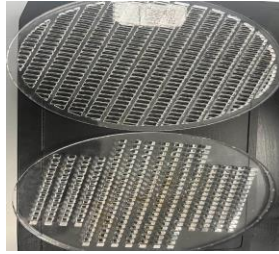


Figure 4

3.4 Panels

To provide a protective casing for the physical prototype, a series of wooden panels were ordered after careful measurement of the frame.

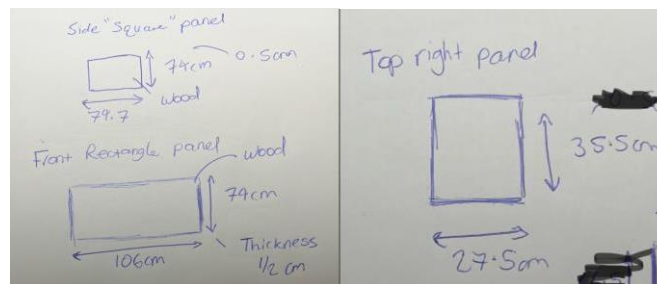


Figure 5

Figure 6

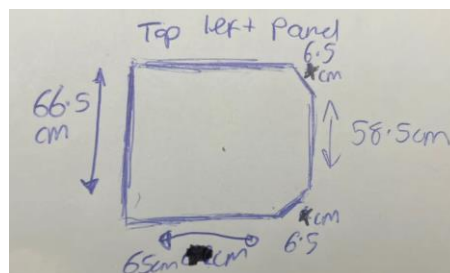


Figure 7

The front rectangular panel was cut to 106 cm by 74 cm with a thickness of 0.5 cm, forming the primary face of the enclosure. On the upper section of the frame, a top right panel was specified at 35.5 cm by 27.5 cm, while the top left panel required more variation due to the irregular shape of the space; here several different lengths were noted, with one of the key dimensions

being 66.5 cm. For the opposite side, a further panel was measured at 74.7 cm by 74 cm, ensuring that the enclosure would align with the overall square footprint of the structure. These panels together offered a lightweight yet practical solution for protecting the system while remaining easy to assemble and remove for maintenance.

4.0 Simulation Context

4.1 Simulation Objectives

To ensure that the reactor was tested adequately, specific objectives were outlined. The end-goal was to ensure that the model and simulation were able to focus on and tackle the following objectives:

Oxygen Purity at Outlet

- ☐ Determine the theoretical value of oxygen concentration purity as a percentage

Breakthrough Behaviour

- ☐ Using data available in previous reports, predict the theoretical time at which nitrogen begins to “breakthrough” the bed during adsorption.

Graphical Representations

- ☐ Present the temperature profile of the reactor ☐
Show pressure drop across the length of the bed.
- ☐ Illustrate breakthrough behavior
- ☐ Display changes in nitrogen and oxygen concentration from inlet ($x = 0$) to outlet ($x = L$).

5.0 Model and Simulation Curation

5.1 Reactor dimensions

As previously mentioned, the reactor refers to each of the adsorption canisters packed with zeolite 13x. These vessels operate as cyclic packed-bed reactors that alternate between adsorbing nitrogen from the compressed air and regenerating through depressurisation and purging. Both adsorption beds are operated out of phase using solenoid valves to ensure almost-continuous oxygen production.

Within the model, each canister can be perceived as a one-dimensional cylindrical reactor operating along the x-axis. A plug flow framework is used alongside the assumption that axial dispersion is present in the canister.

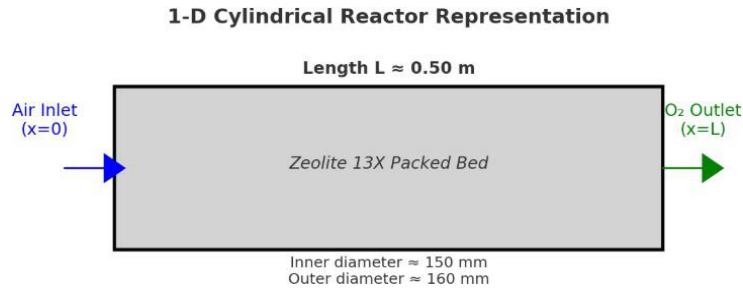


Figure 8

Based on the information noted down on the current reactor, The reactor's geometry and its physical specifications are as follows:

- ☐ Outer diameter (d_o): 160 mm
- ☐ Inner diameter (d_i): ~150 mm (allowing for wall thickness, PVC vessel)
- ☐ Total vessel height: 550 mm
- ☐ Effective packed bed length (L): ~0.50 m (internal volume available for zeolite packing).
- ☐ Wall material: PVC cylinder with CNC-machined aluminium domed ends
- ☐ Sealing: O-ring compression seal at the flanges (8 bolts evenly distributed) .
- ☐ Support/mounting: Each vessel is mounted vertically into the wooden base of the frame via drilled canister holes

5.2 Model Framework for Oxygen purity Concentration

In adsorption-based reactors used for oxygen generators, plug flow with axial dispersion is adopted to simplify the description of transport while retaining the essential physics. The canister is represented as a one-dimensional packed bed along the x-axis; air enters at $x = 0$ and product is withdrawn at $x = L$. The objective is to compute the oxygen concentration purity at the outlet ($x = L$) at the end of the adsorption step. It's worth noting that some equations/symbols will not be in plain text if deemed not to be fit for the format.

State Variables and Closure

To ensure numerical stability, the gas phase is evolved in mole fraction and then converted to absolute concentrations when reporting purity. The state is:

$$[y_{N_2}(x, t), q_{N_2}(x, t), q_{O_2}(x, t), T(x, t)], \quad y_{O_2}(x, t) = 1 - y_{N_2}(x, t).$$

Equation 1

An ideal-gas closure links composition, pressure and temperature to total concentration:

$$C_{\text{tot}}(x, t) = \frac{P(x, t)}{R T(x, t)}, \quad C_i(x, t) = y_i(x, t) C_{\text{tot}}(x, t).$$

Equation 2

A small bound keeps $y_{N_2} \in (10^{-9}, 1 - 10^{-9})$ before evaluating equilibrium to avoid non-physical values.

```
yN2 = y(1:Nx); qN2 = y(Nx+1:2*Nx); qO2 = y(2*Nx+1:3*Nx); T = y(3*Nx+1:4*Nx);
yN2 = max(1e-9, min(1-1e-9, yN2));
yO2 = 1 - yN2;
```

Excerpt 1 — state layout and clamp

Species balances (gas phase)

Axially dispersed plug flow is written directly in mole-fraction form with a sink for solid uptake:

$$\varepsilon C_{\text{tot}}(x, t) \frac{\partial y_{N_2}}{\partial t} + u(x, t) C_{\text{tot}}(x, t) \frac{\partial y_{N_2}}{\partial x} = D_{ax} C_{\text{tot}}(x, t) \frac{\partial^2 y_{N_2}}{\partial x^2} - \frac{1 - \varepsilon}{\varepsilon} \rho_p \frac{\partial q_{N_2}}{\partial t}.$$

Equation 3

```

d2y = zeros(Nx,1);
d2y(2:end-1) = (yN2(3:end) - 2*yN2(2:end-1) + yN2(1:end-2))/dx^2;
dy_dx = zeros(Nx,1);
dy_dx(1) = (yN2(1) - yN2_in)/dx;
dy_dx(2:end) = (yN2(2:end) - yN2(1:end-1))/dx;

dyN2dt = ( Dax.*Ctot.*d2y - U.*Ctot.*dy_dx - ((1-eps)/eps)*rho_p.*dqN2dt ) ./ (eps.*Ctot);

```

Excerpt 2 — discretised y-balance (central dispersion, upwind convection)

Solid uptake (intraparticle kinetics and equilibrium)

A linear driving force (LDF) model represents intraparticle mass transfer:

$$\frac{\partial q_i}{\partial t} = k_{\text{LDF}} (q_i^* - q_i), \quad i \in \{N_2, O_2\}.$$

Equation 4

Equilibrium follows an extended Langmuir isotherm with local partial pressures:

$$q_i^* = \frac{q_{\max,i} b_i P_i}{1 + \sum_j b_j P_j}, \quad P_i = y_i(x, t) P(x, t).$$

Equation 5

```

PN2 = yN2 .* P; P02 = yO2 .* P;
den = 1 + bN2.*PN2 + bO2.*P02;
qN2s = qmaxN2 * (bN2.*PN2) ./ den;
qO2s = qmaxO2 * (bO2.*P02) ./ den;
dqN2dt = kLDF * (qN2s - qN2);
dqO2dt = kLDF * (qO2s - qO2);

```

Excerpt 3 — LDF + extended Langmuir

Momentum (Ergun pressure drop with compressibility)

Pressure varies axially according to the Ergun correlation; the superficial velocity adjusts through molar continuity:

$$-\frac{\partial P}{\partial x} = \frac{150(1-\varepsilon)^2}{\varepsilon^3} \frac{\mu u(x,t)}{d_p^2} + \frac{1.75(1-\varepsilon)}{\varepsilon^3} \frac{\rho_g(x,t) u(x,t)^2}{d_p},$$

Equation 6

$$u(x,t) = \frac{G_m}{C_{\text{tot}}(x,t)}, \quad \rho_g(x,t) = C_{\text{tot}}(x,t) M_{\text{mix}}(x,t), \quad M_{\text{mix}} = y_{N_2} M_{N_2} + y_{O_2} M_{O_2}.$$

Equation 7

The Ergun relation is evaluated algebraically within each RHS call (quasi-steady momentum), which keeps the time integration well-conditioned.

```

Ctot_in = P_in/(R*T(1));
Gm = u_in * Ctot_in;
for i = 1:Nx-1
    Ctot(i) = P(i)/(R*T(i));
    U(i) = Gm / Ctot(i);
    rho_g = Ctot(i) * Mmix(i);
    visc = 150*((1-eps)^2/eps^3) * mu * U(i) / (dp^2);
    inert = 1.75*((1-eps)/eps^3) * rho_g * U(i)^2 / dp;
    dBdx = -(visc + inert);
    P(i+1) = max(1e3, P(i) + dBdx*dx);
end
Ctot(end) = P(end)/(R*T(end)); U(end) = Gm / Ctot(end);

```

Excerpt 4— Ergun/continuity helper

Energy (single temperature for gas + solid)

A lumped energy balance captures axial enthalpy transport and the heat of adsorption:

$$\left(\varepsilon C_{\text{tot}} \bar{C}_{p,g} + (1-\varepsilon) \rho_p C_{p,s} \right) \frac{\partial T}{\partial t} + u C_{\text{tot}} \bar{C}_{p,g} \frac{\partial T}{\partial x} = \lambda_{\text{eff}} \frac{\partial^2 T}{\partial x^2} + (1-\varepsilon) \rho_p \sum_i (-\Delta H_i) \frac{\partial q_i}{\partial t}.$$

Equation 8

```

Ce = eps.*Ctot.*Cp_g_m + (1-eps). *rho_p.*Cp_s;
d2T = zeros(Nx,1);
d2T(2:end-1) = (T(3:end) - 2*T(2:end-1) + T(1:end-2))/dx^2;
dT_dx = zeros(Nx,1);
dT_dx(2:end) = (T(2:end) - T(1:end-1))/dx;

Qads = (1-eps). *rho_p .* ( dH_N2.*dqN2dt + dH_O2.*dqO2dt );
dTdt = ( lambda_eff.*d2T + Qads - (U.*Ctot.*Cp_g_m). *dT_dx ) ./ Ce;

```

Excerpt 5 — temperature balance

Boundary and initial conditions

Inlet at ($x=0$): $y_{N_2} = 0.79$ (air), $P = P_{in}$, $T = T_{in}$.

Outlet ($x = L$): convective outflow for species and energy (zero diffusive gradients).

Initial Fields: $y_{N_2}(x, 0) = 0.79$; $q_i(x, 0) \approx 0$ (regenerated bed); $T(x, 0) = T_{in}$.

```
yN2_0 = yN2_in*ones(Nx,1); qN2_0 = zeros(Nx,1); qO2_0 = zeros(Nx,1);
T0 = T_in*ones(Nx,1);
y0 = [yN2_0; qN2_0; qO2_0; T0];

pars = struct('Nx',Nx,'dx',dx,'eps',eps,'rho_p',rho_p,'R',R, ...
'Dax',Dax,'kLDF',kLDF,'qmaxN2',qmaxN2,'qmaxO2',qmaxO2, ...
'bN2',bN2,'bO2',bO2,'yN2_in',yN2_in, ...
'dp',dp,'mu',mu,'Cp_g_m',Cp_g_m,'Cp_s',Cp_s, ...
'lambda_eff',lambda_eff,'dH_N2',dH_N2,'dH_O2',dH_O2, ...
'P_in',P_in,'u_in',u_in,'T_in',T_in);
```

Excerpt 6 — initialisation & parameter pack

Numerical formulation

The method of lines is used with a uniform grid in x . Axial dispersion is discretised with second-order central differences; convection uses first-order upwind with Dirichlet inlet values for y_{N_2} .

The Ergun pressure and resulting fields $\{P(x), u(x), C_{tot}(x)\}$ are recomputed algebraically inside the RHS using the current $\{y(x), T(x)\}$. Time integration uses a stiff ODE solver with a limited maximum step to resolve the concentration and temperature fronts. The mole-fraction clamp prevents overshoots outside $[0, 1]$.

```
opts = odeset('RelTol',1e-6,'AbsTol',1e-8,'MaxStep',0.2);
[~,Y] = ode15s(@(t,y) rhs_yN2_xT(t,y,pars), [0 tend], y0, opts);
```

Excerpt 7 — solver call

Output metric (reported quantity)

At the end of the adsorption window $t = t_{end}$ the outlet purity is reported in concentration form using local $P(L)$ and $T(L)$:

$$C_{tot}(L) = \frac{P(L)}{RT(L)}, \quad C_{O_2}(L) = y_{O_2}(L) C_{tot}(L), \quad C_{N_2}(L) = y_{N_2}(L) C_{tot}(L),$$

Equation 9

$$\text{Purity}_{O_2}^{(C)}(L) = \frac{C_{O_2}(L)}{C_{O_2}(L) + C_{N_2}(L)} \times 100\%.$$

Equation 10

```
yN2_end = Y(end,1:Nx).';
T_end   = Y(end,3*Nx+1:4*Nx).';
prof_end = compute_pressure_velocity_x(pars, yN2_end, T_end);
Ctot_L = prof_end.Ctot(end);
yO2_out = 1 - yN2_end(end);
C_O2_out = yO2_out * Ctot_L;
C_N2_out = (1 - yO2_out) * Ctot_L;
purityC = 100 * C_O2_out / (C_O2_out + C_N2_out);
fprintf('Concentration-based O2 purity (x=L): %.1f%%\n', purityC);
```

Excerpt 8 — Concentration and purity computation

5.3 Model Framework for Breakthrough Behaviour

The aim here is to determine when nitrogen first “breaks through” at the outlet during a continuous adsorption step, using the same one-dimensional bed model already established for outlet oxygen concentration. The canister is represented along the x-axis with air fed at $x = 0$ and product withdrawn at $x = L$. Breakthrough is reported at two conventional thresholds 10% and 50% of the inlet N_2 fraction (Ruthven, 1984).

Model set-up (kept consistent with the purity framework)

The breakthrough model reuses the governing equations and numerical values from the outlet-purity framework:

- Gas species balance in mole-fraction form, LDF intraparticle kinetics, and the extended Langmuir isotherm are used unchanged; see the previously presented

equations for $y_{N_2}(x, t)$, $q_i(x, t)$, and q_i^* .

- Momentum uses the Ergun correlation evaluated algebraically each time step, with the local superficial velocity obtained from a constant inlet molar-flux anchor via

$$u(x) = G_m / C_{\text{tot}}(x).$$

□ Energy uses the same single-temperature balance for gas and solid with axial conduction and an adsorption-heat source.

□ For numerical robustness, the gas composition is evolved as mole fraction with a small clamp in (0,1); absolute concentrations are recovered where needed from

$$C_{\text{tot}}(x) = P(x)/(RT(x)).$$

Operating conditions and parameters

Measurements from the physical prototype are called upon to tailor the model. Operating pressure at the inlet is kept within the project envelope for adsorption. Typical packed-bed and 13X properties are used where required by the physics:

- Porosity $\varepsilon = 0.37$; particle diameter $d_p = 2$ mm (typical for 13X pellets used in PSA concentrators) (Yang, 1997; Ruthven, 1984).
- Gas viscosity $\mu = 1.8 \times 10^{-5}$ (unit = Pascal·seconds) for air at ambient.
- Axial dispersion $D_{ax} = 7.5 \times 10^{-5}$ m²/s, within the typical packed-bed range values (Levenspiel, 1999; Ruthven, 1984).
- LDF rate $kLDF = 0.11$ s⁻¹, representative of the tiny-pellet beds used within the medical oxygen industry.
- Isotherm (13X air):
 $q_{\text{max},N_2} = 1.8$ (unit = mol per kilo),
 $b_{N_2} = 0.07$ bar⁻¹,
 $q_{\text{max},O_2} = 0.45$ (unit = mol per kilo),
 $b_{O_2} = 0.02$ bar⁻¹ (Chihara & Suzuki, 1983; Yang, 1997).
- Thermophysical: $C_{p,g} = 29$ J·mol⁻¹·K⁻¹ (air, molar basis), $C_{p,s} = 900$ J·kg⁻¹·K⁻¹,
 $\lambda_{\text{eff}} = 0.15$ W·m⁻¹·K⁻¹, adsorption heats: $\Delta H_{N_2} \approx 20$ kJ·mol⁻¹,
 $\Delta H_{O_2} \approx 12$ kJ·mol⁻¹.

```
u_in      = 0.09;
Dax       = 7.5e-5;
kLDF      = 0.11;
mu        = 1.8e-5;
lambda_eff = 0.15;
Cp_g_m    = 29.0;
Cp_s      = 900.0;
dH_N2     = 2.0e4;
dH_O2     = 1.2e4;
```

Excerpt 9 – Specific Values

Breakthrough definitions and outputs

Let the inlet nitrogen fraction be $y_{N_2,\text{in}} = 0.79$. Define the 10% breakthrough time and the 50% breakthrough time as $t_{B,10}$ and $t_{B,50}$ respectively by the first instances when:

$$y_{N_2}(x=L, t) \geq 0.10 y_{N_2,\text{in}} \quad \text{and} \quad y_{N_2}(x=L, t) \geq 0.50 y_{N_2,\text{in}}$$

Equation 11

5.5 Model Framework for Graphical Representation

This framework aims to develop a compact and defensible model that can graphically show the reactor's temperature profile, the pressure drop along the bed, the nitrogen and oxygen composition from inlet $x=0$ to outlet $x=L$ and finally the breakthrough behaviour at the outlet.

Equation and stabilisation

To ensure numerical clarity and robustness, the gas phase is advanced in mole fractions converted to absolute concentrations only when reporting purity. The state is shown below.

$$\mathbf{z}(x, t) = [y_{N_2}(x, t), q_{N_2}(x, t), q_{O_2}(x, t), T(x, t)],$$

Equation 12

$$y_{O_2}(x, t) = 1 - y_{N_2}(x, t).$$

Equation 13

Before evaluating equilibrium and kinetic terms, a light regularisation prevents non-physical values at machine precision:

$$y_{N_2} \leftarrow \min(1 - \varepsilon_y, \max(\varepsilon_y, y_{N_2})), \quad \varepsilon_y = 10^{-9}$$

Equation 14

This clipping leaves the resolved solution unchanged to numerical accuracy while avoiding singular denominators in the extended-Langmuir isotherm.

Governing Equations

The graphical representation framework reuses the same governing equations, closures, and numerical values as the outlet-purity model. Gas transport is advanced in mole-fraction form with LDF intraparticle kinetics and the extended Langmuir equilibrium unchanged from the earlier formulation (see Governing Equations). Momentum is evaluated algebraically at each time step via the Ergun correlation, with the local superficial velocity obtained from a constant inlet molar-flux anchor. Thermal behaviour follows the single-temperature gas–solid energy balance with axial conduction and an adsorption-heat source, as previously specified. For numerical robustness, composition is evolved as a mole fraction with a light clamp in (0,1). Absolute concentrations are only recovered where required using the ideal-gas closure.

```
PN2 = yN2 .* P; P02 = (1 - yN2) .* P;  
den = 1 + bN2.*PN2 + bO2.*P02;  
qN2s = qmaxN2 * (bN2.*PN2) ./ den;  
qO2s = qmaxO2 * (bO2.*P02) ./ den;  
dqN2dt = kLDF * (qN2s - qN2);  
dqO2dt = kLDF * (qO2s - qO2);
```

Excerpt 10 - LDF

Boundary conditions, initialisation, and outputs

These conditions are in place to aid in the initialisation and stabilisation of the framework

- Domain: one-dimensional bed, $L=0.55$ m, inlet at $x=0$, outlet at $x=L$.
- Inlet composition: $y_{N_2}(0, t) = 0.79$ (air), hence $y_{O_2}(0, t) = 0.21$.
- Convection uses a one-sided gradient at the inlet
- Thermal: zero axial conductive heat flux at both ends; the feed temperature enters through convection in the energy equation.
- $T(x)$ profiles used to present the temperature field;
- $P(x)$ to report the pressure drop

□ $y_{N_2}(x)$ and $y_{O_2}(x)$ are at representative times

6.0 Results, Analysis and Discussion

6.1 Section Context

A complete dissection of the results given from each of the modelling frameworks will be implemented within this section. Results, subsequent analysis and detailed discussions of potential improvements will be carried out in this section.

6.2 Oxygen Purity Concentration Results

Running the adsorption-only script returns a single snapshot at the product end of the bed. At time 18 seconds and position x equals L , the outlet oxygen concentration is 129.94 mol per cubic metre and the outlet nitrogen concentration is 50.14 mol per cubic metre. On that basis, the concentration-defined oxygen purity at the outlet is 72.2 percent. This is a point estimate at one location and one time. It is not a flow-weighted product purity over a dedicated production window and it is not a cycle-averaged figure at cyclic steady state. In other words, it is a conservative picture of the column at the end of adsorption, before any benefit from blowdown, purge, or pressure equalisation is taken into account.

The headline number at 72.2 percent sits below the medical oxygen threshold used in international guidance, where oxygen concentrators are expected to deliver at least 82 percent oxygen at the rated flow, with most devices designed to operate in the band from roughly 82 to 96 percent depending on flow and conditions (WHO, 2015; ISO, 2014; ISO, 2020).

6.3 Concentration/Graph Analysis

The purity landed where it did because of a combination of step definition, transport, thermodynamics and hydrodynamics embedded in the code. The model samples the state at the end of an adsorption step that lasts 18 seconds. There is no cycle selection of a “produce” sub-window; everything is assessed at the instant the step ends. In a real pressure-swing cycle,

nitrogen loaded during pressurisation and adsorption is expelled during blowdown and purge, and purity is calculated only across the chosen production interval. That difference in definition alone can shift an apparent 70-something into the mid-80s once timing and cut are tuned in a full cycle simulator (Santos et al., 2006; UNICEF, 2022).

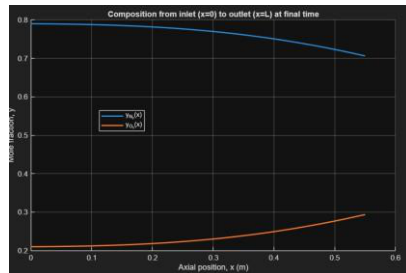


Figure 9

Mass transfer and dispersion also earn their say. The bed uses pellets of 2 millimetres with porosity 0.37 and an axial dispersion coefficient of $7.5\text{e-}5$ square metres per second at a superficial velocity of 0.09 metres per second. Those settings produce a moderately broad mass-transfer zone for nitrogen. A broader front arrives earlier and with more smear at the outlet, so more nitrogen slips through by the time the step ends, and purity drops accordingly.

Literature on small medical oxygen concentrators repeatedly shows that moving to smaller particles, or lowering dispersion by design, sharpens the front and improves the end-of-step oxygen fraction for a given bed length and timing (Carty and Bilodeau, 2023; Arora et al., 2021).

Heat effects compound the issue. The code includes a single-temperature energy balance and an adsorption heat source using 20,000 joules per mole for nitrogen and 12,000 joules per mole for oxygen. Exotherms raise the gas–solid temperature as the front passes. In reality, equilibrium constants fall with temperature; if that temperature dependence is not modelled explicitly, the model can over-predict working capacity upstream and then still show a purity penalty

downstream because the front simply moves faster. Contemporary assessments of medical oxygen cycles emphasise that temperature-dependent adsorption equilibria and, ideally, a two-temperature energy model are needed to quantify these effects cleanly (Carty and Bilodeau, 2023).

Hydrodynamics matter as well. The Ergun relation is evaluated with a constant inlet molar-flux anchor. As pressure and therefore total molar concentration fall along the bed, the computed

superficial velocity increases. That accelerates the approach of the nitrogen front to the outlet and again nudges purity downward at the end-of-step cut. Small medical concentrators are usually run under compressor control closer to a fixed volumetric intake at the device boundary, and practical systems also exchange gas between beds through pressure equalisation lines.

Both features change the velocity–pressure profile and, once included in the cycle, often recover purity for the same bed and material (Santos et al., 2006; ISO, 2014).

Parameter provenance is the final driver. The extended Langmuir parameters and the single LDF rate constant at 0.11 per second are reasonable for a sodium-form 13X but they are not tuned to a specific sieve batch, temperature program, or particle geometry. Small shifts in the affinity parameter for nitrogen or in the LDF rate will reposition the mass-transfer zone and change outlet purity by several points at fixed timing. Best-practice modelling in this space fits isotherms versus temperature and validates overall kinetics against step tests before drawing cycle-level conclusions (Santos et al., 2006; Carty and Bilodeau, 2023).

Set against clinical practice, the number at 72.2 percent has two immediate implications. First, it is below the 82 percent benchmark that standards and procurement use to define medical-grade output for concentrators at their rated flow (WHO, 2015; ISO, 2014; UNICEF, 2020). Second, if the downstream device is a Venturi mask, those nozzles are calibrated to deliver fixed oxygen fractions such as 24, 28, 35, 40 or 60 percent from a high-concentration oxygen feed.

Feeding a Venturi with 72 percent oxygen will under-deliver the number printed on the nozzle unless the setup is re-rated, because the Venturi relies on controlled entrainment of room air into an oxygen-rich jet (NHS GGC, 2022; O'Driscoll et al., 2017).

6.4 Improvements and Sensitivity

The snapshot purity at 72.2 percent is useful as a stress test, but the performance needs to be reported in the same currency that clinicians and standards use. The next step is to simulate full cycles and report flow-weighted product purity, recovery and productivity over the chosen production window at cyclic steady state. That will align results with the ≥ 82 percent medical threshold and with how regulators and buyers assess concentrators (WHO, 2015; ISO, 2014).

Two physics upgrades will make those cycle results credible. First, introduce temperature-dependent equilibria and a two-temperature energy model with interphase heat transfer so the

local hot spots and the resulting loss in working capacity are captured honestly. This usually lowers optimistic end-of-step numbers in single-temperature models but gives the right levers to recover purity through timings and purge (Carty and Bilodeau, 2023). Second, revisit the hydrodynamic closure. Replace the constant molar-flux anchor with a fixed volumetric inflow representative of the compressor and add pressure-equalisation between beds. Together with a modest increase in purge ratio and a slightly shorter adsorption cut, this is often enough to push small units from the high-70s into the high-80s or low-90s without hardware changes (Santos et al., 2006).

Material and geometry choices remain powerful dials. A careful reduction in particle size from 2 millimetres toward 1–1.5 millimetres, or a small increase in bed length coupled with a lower superficial velocity at the same inlet pressure, will sharpen the mass-transfer zone and delay slip; any added pressure drop can be traded against compressor headroom. In parallel, fit the extended-Langmuir and LDF parameters to batch data so cycle optimisation rests on validated equilibria and kinetics. Finally, if the product feeds a Venturi delivery system, either target outlet purity in the high-80s to low-90s so the nozzle chart remains valid, or re-rate the FiO_2 tables explicitly against a sub-90 percent source to avoid under-delivery at the bedside (O'Driscoll et al., 2017).

6.5 Breakthrough Behaviour Results

Running the breakthrough script with the regenerated-bed initial condition returns a clean set of outlet diagnostics. The outlet nitrogen mole-fraction trace spans approximately 0.000 to 0.706 over the simulated window, which is consistent with a bed that starts free of nitrogen and then admits a nitrogen-bearing feed until the front arrives at the product end. Using the standard first-crossing definition, breakthrough at 10 percent of the feed nitrogen mole fraction occurs at 6.87 seconds. The mid-breakthrough point at 50 percent of the feed nitrogen mole fraction occurs at 27.24 seconds. The mass-transfer-zone estimate derived from the 10 to 90 percent width evaluated at the 50 percent time is returned as 0.000 metres by the current interpolation routine. At the final simulated time the outlet oxygen concentration is 52.06 mol per cubic metre and the concentration-based oxygen purity computed from the local total concentration is 29.4 percent.

Two contextual points are important. First, the initial condition is deliberately optimistic for a real cycle because the bed gas is set to nitrogen-free at time zero. Second, the purity printed at the

end of the run is a simple end-of-window snapshot rather than a flow-weighted value over a defined production interval and it occurs long after significant nitrogen slip is present at the outlet. In short, the headline quantities to pay attention to here are the 10 percent and 50 percent breakthrough times and what they imply for practical step selection rather than the late-time purity. The 10 percent and 50 percent thresholds are conventional in fixed-bed analysis because they bracket the onset of slip and the middle of the front, and are widely used to compare front velocities across conditions and materials (Knox, 2016).

6.6 Breakthrough/Graph Analysis

Read alongside the group technical report and Lorenzo's draft, the numbers sit where the physics says they should for this scale and set of inputs. The bed length is 0.55 metres, pellets are 2 millimetres, porosity is 0.37, and the superficial velocity at the inlet is 0.09 metres per second. Axial dispersion is set at $7.5\text{e-}5$ square metres per second and the linear driving force rate constant is 0.11 per second.

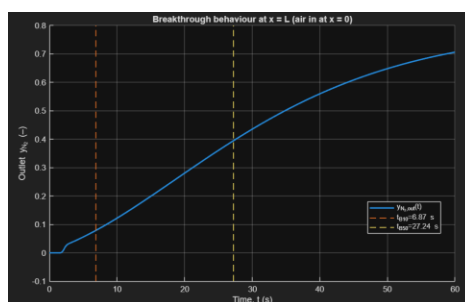


Figure 10

Those values give a moderately broad mass-transfer zone. With a regenerated gas phase at time zero, the front needs a few seconds to establish. That is why the first detectable slip at 10 percent of the inlet nitrogen content arrives at 6.87 seconds. The mid-front arrival at 27.24 seconds signals that, by around half a minute, the front has propagated almost the full length and the outlet is no longer suitable for collection without recycling or cutting the step. This basic story is consistent with canonical fixed-bed behaviour, where the front propagates as a wave whose width is governed by film and intraparticle mass transfer as well as axial dispersion, and

whose speed is set by the interplay of convection and equilibrium uptake (Ruthven et al., 1994; Knox, 2016).

The reported mass-transfer-zone length of 0.000 metres at the 50 percent time is a numerical artefact rather than a physical conclusion. The routine computes the 10 to 90 percent width from a single time slice by interpolating the normalised nitrogen profile against position. In this run the profile at the chosen time does not cleanly bracket both 10 and 90 percent levels on distinct grid points, so the linear interpolation collapses to a zero width. In practice, medical oxygen beds of this scale and pellet size exhibit finite zone widths that are a material fraction of the bed length; the literature usually quantifies this using a 10 to 90 percent distance or an equivalent slope-based measure (Knaebel, 2001; NPTEL MTZ notes). The zero returned here therefore indicates the interpolation needs to be made more robust and the spatial grid likely needs to be refined.

What changes the breakthrough times is well documented and the present model exposes most of the levers. Increasing bed length while holding velocity and all else constant increases both 10 percent and 50 percent breakthrough times more or less linearly with length because the front has farther to travel. Reducing superficial velocity by throttling the intake or increasing the pressure drop moves the front more slowly, increasing breakthrough times but at the cost of feed throughput. Decreasing particle size from 2 millimetres toward 1 to 1.5 millimetres reduces intraparticle diffusion path and steepens the front, which tends to delay 10 percent crossing for a given length, although the pressure-drop penalty must be absorbed by the compressor.

Strengthening nitrogen equilibrium capacity or selectivity through higher affinity or a lithium-exchanged sieve pushes breakthrough later. Heat effects pull in the opposite direction: exotherms raise the local temperature and, for most zeolites, reduce effective affinity at the hot spot, advancing the front. All of these relationships are consistent with contemporary medical oxygen reviews, where faster cycles are enabled by small particles and short beds but must be balanced against compressor headroom and thermal management (Arora et al., 2021; Chai et al., 2011).

It is also worth assessing the times against what is typically used in concentrators. Many commercial two-bed units operate with total cycle times near 10 seconds per bed pair in mature designs, with the adsorption cut occupying only a fraction of that window, while rapid variants push total cycles down to a few seconds using very small particles and aggressive equalisation

(Miller, 1991; Arora et al., 2021; Chai et al., 2011). Against that backdrop, a 10 percent breakthrough at 6.87 seconds suggests that a conservative produce window might sensibly be limited to the first several seconds of adsorption at these conditions, unless purge and equalisation strategies are introduced to reshape the profiles. The 50 percent value at 27.24 seconds is too late to be useful in a medical cycle because product purity would already be compromised well before that point. These comparisons align with the medical oxygen standards environment, where the goal is to maintain at least 82 percent oxygen during production at the device's rated flow; late-time operation with significant nitrogen slip will not satisfy that bar (ISO, 2014; WHO, 2015).

6.7 Discussion and Refinement

The breakthrough timings tell a clear operational story: at these settings the only defensible produce window is the early part of adsorption, well before the 10 percent crossing at 6.87 seconds; once the outlet rises, purity deteriorates quickly. To translate this into device-relevant performance, the model should step up from single-pass breakthrough to a full two-bed cycle run to cyclic steady state, and report flow-weighted product purity, recovery and productivity over the selected produce cut, which is the currency used in ISO 80601-2-69 and WHO guidance for medical concentrators (ISO, 2014; WHO, 2015). Two physics upgrades have the highest return. First, bring temperature dependence into the equilibrium via van't Hoff-type affinity terms and move to a two-temperature energy model with interphase heat transfer; this captures hot-spot penalties on capacity and yields trustworthy front speeds for optimisation (Carty and Bilodeau, 2023). Second, revise the hydrodynamic closure to a fixed inlet volumetric flow consistent with the compressor and add pressure-equalisation between beds; together with modest increases in purge ratio and a slightly shorter adsorption cut, this routinely shifts small units from high-70s into high-80s or low-90s cycle purities (Santos et al., 2006).

Make the MTZ estimate robust by refining the axial grid beyond 80 nodes and using a monotone interpolant or a gradient/integral definition so zero-width artefacts disappear. Finally, explore practical levers under compressor headroom: reduce pellet size from 2 mm toward 1–1.5 mm, lengthen the bed by roughly 10–20 percent, or trim superficial velocity by 10–20 percent; each steepens the front and delays breakthrough, creating a longer high-purity window (Chai et al., 2011; Arora et al., 2021).

6.8 Pressure Drop and Temperature Profile

Running the visualisation framework on the breakthrough initial condition produces coherent thermal, hydraulic and compositional pictures across the bed.

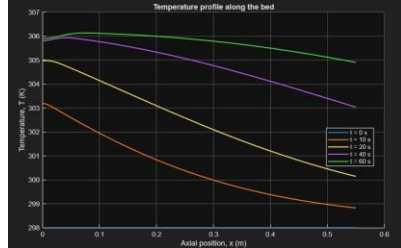


Figure 11

The temperature plot shows an immediate and localised rise near the inlet as adsorption starts, with the bed warming from the 298 feed to roughly 303 at 10 seconds, 305 at 20 seconds and just over 306 by 40 to 60 seconds. The axial temperature gradient is gentle and negative from inlet to outlet at each time, which is what would be expected when the heat source is strongest where the nitrogen front first engages the sorbent while axial conduction and convective carriage spread that heat downstream.

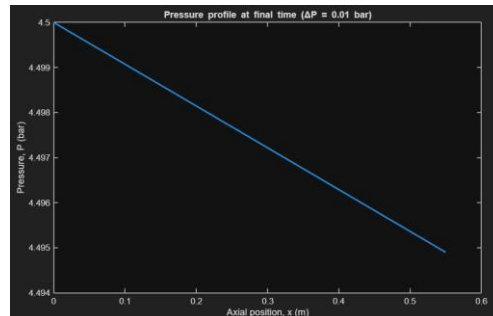


Figure 12

The pressure plot at the final time is close to linear with a total drop of about 0.01 bar from 4.50 bar at the inlet to 4.49 bar at the outlet, consistent with the modest superficial velocity and the relatively large 2 millimetre pellets. The code also shows, at the same settings, a nitrogen breakthrough at the outlet with the first crossing at 10 percent of the inlet nitrogen mole fraction

at 6.87 seconds and the mid-breakthrough at 27.24 seconds; those timings are helpful for interpreting the temperature curves because the strongest temperature rise sits ahead of and around the front, then relaxes as the zone passes. The plots therefore give a compact, time-resolved view of how heat release, dispersion and pressure loss play together in this small 13X bed under an adsorption step, using the same 1-D physics and boundary conditions defined in the model pack for species, energy and Ergun momentum with a constant inlet molar-flux anchor.

6.9 Pressure Drop and Temperature Profile Analysis

The temperature profiles tell the expected story for a nitrogen-selective zeolite. Adsorption is exothermic, with typical integral heats for nitrogen on sodium-form or lithium-exchanged low silica X in the ballpark of 20 to 35 kilojoules per mole depending on loading and cation form. A local rise of about 7 to 8 kelvin in this bed is therefore entirely plausible at these pressures and step lengths, and the pattern of a peak near the inlet that slowly diminishes downstream mirrors textbook 1-D predictions when axial dispersion and axial thermal conduction are included in a single-temperature model. The mild axial gradient in each time slice is consistent with the relatively high effective axial conductivity used in the framework and with the fact that the heat source term decays as the uptake rate falls once the mass-transfer zone moves away from the inlet. These signatures align with the classical treatment of fixed-bed thermal waves in adsorption columns and with more modern 1-D simulations for small medical oxygen beds that report modest but non-negligible temperature rises during adsorption stages. (Ruthven, 1984; Jumaeva et al., 2021; Chai et al., 2011).

The pressure plot also behaves as it should. With porosity 0.37, particle diameter 2 millimetres, viscosity roughly $1.8\text{e-}5$ pascal-seconds and superficial velocity 0.09 metres per second, the Ergun correlation predicts a small frictional loss with a near-linear profile when evaluated at this scale. A drop of about 0.01 bar across 0.55 metres falls squarely within that regime and indicates that pressure loss is not the limiting factor in this operating window; the compressor is not paying a heavy penalty to move gas through the bed at these flow conditions. The constant-molar-flux closure means the local superficial velocity increases slightly as total concentration falls downstream, which will convect the mass-transfer zone marginally faster near the outlet; this is visible indirectly in the way the downstream curves are a little cooler than the inlet region at a given time. The treatment is consistent with the Ergun-based momentum statement and with

standard design guidance for packed beds at moderate Reynolds numbers. (Ergun, 1952; Hassan et al., 2024; model equations pack).

Taken together, the temperature and pressure trends align with what is described in the internal model documentation and with wider 1-D literature. The bed warms rapidly where the nitrogen front forms, settles toward a gentle gradient as the uptake rate wanes, and experiences only a light pressure burden. That match gives confidence that the discretisation and closures are behaving as intended. It also lines up with the system-level direction in the group technical report and in Lorenzo's draft, where temperature sensing is flagged as a safety and performance requirement and where Venturi delivery is considered as a downstream option. If a Venturi device is used, thermal stability and predictable outlet composition during the produce window matter because the nozzle charts assume a stable oxygen-rich source; the present thermal profiles suggest the bed is not experiencing excessive hot spots that would destabilise operation at this scale.

6.9.1 Discussion and Future Improvement

The figures confirm the framework is capturing the right first-order physics for a small 13X bed under adsorption, but the results should now be translated into device-relevant performance. Priority one is to make the thermal–equilibrium link explicit. The observed 7–8 K rise needs to feed into temperature-dependent equilibria and, ideally, a two-temperature energy model with interphase heat transfer. This closes the loop between heat release and working capacity, giving trustworthy front speeds and a realistic purity penalty to optimise against (Ruthven, 1984; Carty and Bilodeau, 2023).

Reporting should then move from single-pass snapshots to full two-bed cycles at cyclic steady state, with flow-weighted product purity, recovery and productivity over an explicitly selected produce window. Those are the metrics used in practice and they are already defined in the project's equations pack, so they can be adopted with minor boundary-switching extensions to the present code.

A short sensitivity sweep will make the trade-offs tangible. Reducing pellet size from 2 mm toward 1–1.5 mm, trimming the inlet superficial velocity from 0.09 m s⁻¹ by 10–20 percent, or lengthening the 0.55 m bed by roughly 10–20 percent should steepen the mass-transfer zone,

delay nitrogen slip and concentrate heat upstream, thereby extending the high-purity window for a given pressure budget (Chai et al., 2011; Ruthven, 1984).

Finally, validate the hydraulic model in hardware. Compare the Ergun-predicted drop with pressure-tap data and adjust effective porosity or sphericity if needed; the literature shows packed-bed pressure loss is sensitive to particle shape and voidage, and classic correlations can misestimate without calibration (Allen and Di Felice, 2013; Hassan et al., 2024).

7.0 Conclusion

7.1 Final Thoughts

Taken together, the modelling paints a consistent picture of the prototype 13X bed and what must change to reach medical-grade output. The adsorption-only purity snapshot at the product end returns 72.2 percent at 18 seconds, while the breakthrough test from a regenerated gas phase gives a first slip at 10 percent of the inlet nitrogen at 6.87 seconds and a mid-breakthrough at 27.24 seconds. Over the same conditions the bed warms from 298 to about 306, with a gentle inlet-to-outlet gradient that tracks the exothermic front, and the pressure loss is small at roughly 0.01 bar across 0.55 metres. These signatures are exactly what a 1-D, nitrogen-selective zeolite bed should show: hydraulically easy to drive, modestly exothermic, and purity-limited by early nitrogen slip when product is taken late in the step. The analysis explains why. The metrics used are point snapshots rather than cycle-level, flow-weighted figures; the mass-transfer zone is broadened by 2 millimetre pellets and axial dispersion; the constant molar-flux closure slightly accelerates the front downstream; and the single-temperature energy balance does not yet translate the 7–8 kelvin rise into a reduction in equilibrium capacity. As a result, late-time purity and the 29.4 percent end-of-window figure in the breakthrough run cannot be used to judge clinical performance.

The path forward is therefore clear. Shift to full two-bed cycle simulations at cyclic steady state and report flow-weighted purity, recovery and productivity over a defined produce window.

Introduce temperature-dependent equilibria and, ideally, a two-temperature energy model so heat release and working capacity are coupled properly. Make the mass-transfer-zone measure robust with a finer axial grid and stable interpolation. Then use standard levers to extend the high-purity window without radical hardware change: pressure equalisation and timing

optimisation, modest adjustments to particle size, superficial velocity and bed length, and parameter fitting against isotherm and step-test data. If the unit is intended to feed Venturi delivery, target outlet purity in the high 80s to low 90s during production or re-rate downstream FiO₂ tables accordingly. With those changes in place, the same platform can credibly reach the ≥ 82 percent band expected of medical concentrators.

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Appendix

Oxygen Purity Code

```
function Oxygen_Purity_Test
```

```
R      = 8.314;
T_in   = 298.0;
P_bar  = 4.5;
P_in   = P_bar*1e5;
yN2_in = 0.79;
```

```
L      = 0.55;
eps    = 0.37;
rho_p  = 1100.0;
dp     = 2e-3;
```

```
u_in   = 0.09;
Dax    = 7.5e-5;
kLDF   = 0.11;
mu     = 1.8e-5;
lambda_eff = 0.15;
Cp_g_m = 29.0;
Cp_s   = 900.0;
dH_N2  = 2.8e4;
dH_O2  = 1.2e4;
```

```
qmaxN2 = 1.8;
bN2bar = 0.07;
qmaxO2 = 0.45;
bO2bar = 0.02;
bN2    = bN2bar/1e5;
bO2    = bO2bar/1e5;
```

```
Nx     = 80;
x       = linspace(0,L,Nx).';
dx     = L/(Nx-1);
tend    = 18.0;
```

```
yN2_0 = yN2_in*ones(Nx,1);
qN2_0 = zeros(Nx,1);
qO2_0 = zeros(Nx,1);
T0     = T_in*ones(Nx,1);
y0     = [yN2_0; qN2_0; qO2_0; T0];
```

```
pars = struct('Nx',Nx,'dx',dx,'eps',eps,'rho_p',rho_p,'R',R, ...
'Dax',Dax,'kLDF',kLDF,'qmaxN2',qmaxN2,'qmaxO2',qmaxO2, ...
'bN2',bN2,'bO2',bO2,'yN2_in',yN2_in, ...
'dp',dp,'mu',mu,'Cp_g_m',Cp_g_m,'Cp_s',Cp_s, ...
'lambda_eff',lambda_eff,'dH_N2',dH_N2,'dH_O2',dH_O2, ...
'P_in',P_in,'u_in',u_in,'T_in',T_in);
```

```
opts = odeset('RelTol',1e-6,'AbsTol',1e-8,'MaxStep',0.2);
[~,Y] = ode15s(@rhs_yN2_x1(t,y,pars), [0 tend], y0, opts);
```

```
yN2_end = Y(end,1:Nx).';
T_end   = Y(end,3*Nx+1:4*Nx).';
```

```
prof_end = compute_pressure_velocity_x(pars, yN2_end, T_end);
Ctot_L = prof_end.Ctot(end);
yO2_out = 1 - yN2_end(end);
C_O2_out = yO2_out * Ctot_L;
C_N2_out = (1 - yO2_out) * Ctot_L;
purityC = 100 * C_O2_out / (C_O2_out + C_N2_out);
```

```
fprintf('Outlet O2 concentration (x=L): %.2f mol/m^3\n', C_O2_out);
fprintf('Outlet N2 concentration (x=L): %.2f mol/m^3\n', C_N2_out);
fprintf('Concentration-based O2 purity (x=L): %.1f%%\n', purityC);
```

```
end
```

```
function dy = rhs_yN2_x1(~,y,par)
```

```
Nx = par.Nx; dx = par.dx; eps = par.eps; rho_p = par.rho_p;
kLDF = par.kLDF; R = par.R;
qmaxN2 = par.qmaxN2; qmaxO2 = par.qmaxO2; bN2 = par.bN2; bO2 = par.bO2;
Dax = par.Dax; Cp_g_m = par.Cp_g_m; Cp_s = par.Cp_s; lambda_eff = par.lambda_eff;
dH_N2 = par.dH_N2; dH_O2 = par.dH_O2; yN2_in = par.yN2_in;
```

```
yN2 = y(1:Nx);
qN2 = y(Nx+1:2*Nx);
qO2 = y(2*Nx+1:3*Nx);
T = y(3*Nx+1:4*Nx);
```

```
prof = compute_pressure_velocity_x(par, yN2, T);
P = prof.P; U = prof.U; Ctot = prof.Ctot;
```

```

yN2 = max(1e-9, min(1-1e-9, yN2)); yO2 = 1 - yN2;

P02 = yN2 .* P; P02 = yO2 .* P;
den = 1 + hN2.*P02 + hO2.*P02;
qN2s = qmaxN2 * (hN2.*P02) ./ den;
qO2s = qmaxO2 * (hO2.*P02) ./ den;

dqN2dt = k1DF * (qN2s - qN2);
dqO2dt = k1DF * (qO2s - qO2);

dZy = zeros(Nx,1);
dZy(2:end-1) = (yN2(3:end) - 2*yN2(2:end-1) + yN2(1:end-2))/dx^2;
dZy(1) = (yN2(2) - yN2(1))/dx^2;
dZy(end) = (yN2(end-1) - yN2(end))/dx^2;

dy_dx = zeros(Nx,1);
dy_dx(1) = (yN2(1) - yN2_in)/dx;
dy_dx(2:end) = (yN2(2:end) - yN2(1:end-1))/dx;

dyN2dt = ( Dax.*Ctot .* dZy - U.*Ctot .* dy_dx - ((1-eps)/eps)*rho_p.*dqN2dt ) ./ (eps.*Ctot);

Ce = eps.*Ctot.*Cp_g_m + (1-eps).*rho_p.*Cp_s;

dT = zeros(Nx,1);
dT(2:end-1) = (T(3:end) - 2*T(2:end-1) + T(1:end-2))/dx^2;
dT(1) = (T(2) - T(1))/dx^2;
dT(end) = (T(end-1) - T(end))/dx^2;

dT_dx = zeros(Nx,1);
dT_dx(1) = (T(1) - T(1))/dx;
dT_dx(2:end) = (T(2:end) - T(1:end-1))/dx;

Qads = (1-eps).*rho_p .* ( dH_N2.*dqN2dt + dH_O2.*dqO2dt );

dTdt = ( lambda_eff.*dT + Qads - (U.*Ctot.*Cp_g_m).*dT_dx ) ./ Ce;

dy = [dyN2dt; dqN2dt; dqO2dt; dTdt];
end

```

```

function prof = compute_pressure_velocity_x(par, yN2, T)
Nx = par.Nx; dx = par.dx; R = par.R; eps = par.eps;
dp = par.dp; mu = par.mu; P_in = par.P_in; u_in = par.u_in;

yN2 = max(1e-9, min(1-1e-9, yN2)); yO2 = 1 - yN2;
M_N2 = 0.0280134; M_O2 = 0.031998;
Mmix = yN2*M_N2 + yO2*M_O2;

Ctot_in = P_in/(R*T(1));
Gm = u_in * Ctot_in;

P = zeros(Nx,1);
U = zeros(Nx,1);
Ctt = zeros(Nx,1);
P(1) = P_in;

for i = 1:Nx-1
    Ctt(i) = P(i)/(R*T(i));
    U(i) = Gm / Ctt(i);
    rho_g = Ctt(i) * Mmix(i);

    visc = 150*((1-eps)^2/eps^3) * mu * U(i) / (dp^2);
    inert = 1.75*((1-eps)/eps^3) * rho_g * U(i)^2 / dp;
    dPdx = -(visc + inert);

    P(i+1) = max(1e3, P(i) + dPdx*dx);
end

Ctt(end) = P(end)/(R*T(end));
U(end) = Gm / Ctt(end);

prof = struct('P',P,'U',U,'Ctot',Ctt);
end

```

Breakthrough Behaviour Code

```

function Breakthrough_Time_Test
    R      = 8.314;
    T_in   = 298.0;
    Pbar   = 4.5;
    P_in   = Pbar*1e5;
    yN2_in = 0.79;

    L      = 0.55;
    eps    = 0.37;
    rho_p  = 1100.0;
    dp     = 2e-3;

    u_in   = 0.09;
    Dax    = 7.5e-5;
    kLDF   = 0.11;
    mu     = 1.8e-5;
    lambda_eff = 0.15;
    Cp_g_m = 29.0;
    Cp_s   = 900.0;
    dH_N2  = 2.0e4;
    dH_O2  = 1.2e4;

    qmaxN2 = 1.8;    bN2bar = 0.07;
    qmaxO2 = 0.45;   bO2bar = 0.02;
    bN2     = bN2bar/1e5;    bO2 = bO2bar/1e5;

    Nx      = 80;
    x        = linspace(0,L,Nx).'; %ok<NASGU>
    dx       = L/(Nx-1);
    tend     = 60.0;

    yN2_0   = zeros(Nx,1);
    qN2_0   = zeros(Nx,1);

```

```

qO2_0 = zeros(Nx,1);
T0     = T_in*ones(Nx,1);
y0     = [yN2_0; qN2_0; qO2_0; T0];

pars = struct('Nx',Nx,'dx',dx,'eps',eps,'rho_p',rho_p,'R',R, ...
    'Dax',Dax,'kLDF',kLDF,'qmaxN2',qmaxN2,'qmaxO2',qmaxO2, ...
    'bN2',bN2,'bO2',bO2,'yN2_in',yN2_in, ...
    'dp',dp,'mu',mu,'Cp_g_m',Cp_g_m,'Cp_s',Cp_s, ...
    'lambda_eff',lambda_eff,'dH_N2',dH_N2,'dH_O2',dH_O2, ...
    'P_in',P_in,'u_in',u_in,'T_in',T_in);

opts = odeset('RelTol',1e-6,'AbsTol',1e-8,'MaxStep',0.25);
[Tsol, Y] = ode15s(@(t,y) rhs_yN2_xT(t,y,pars), [0 tend], y0, opts);

yN2_out_trace = Y(:, 1:pars.Nx);
yN2_out_trace = yN2_out_trace(:, end);
thr10 = 0.10*yN2_in; thr50 = 0.50*yN2_in;

tB10 = localFirstCross(yN2_out_trace, Tsol, thr10);
tB50 = localFirstCross(yN2_out_trace, Tsol, thr50);

MTZ = NaN;
if ~isnan(tB50)
    [~,k] = min(abs(Tsol - tB50));
    yprof = (Y(k,1:Nx)).'/yN2_in;
    xgrid = linspace(0,L,Nx).';
    x10 = interp1(yprof, xgrid, 0.10, 'linear', 'extrap');
    x90 = interp1(yprof, xgrid, 0.90, 'linear', 'extrap');
    MTZ = max(0, x90 - x10);
end

fprintf('Outlet yN2 range: [%3f, %3f]\n', min(yN2_out_trace), max(yN2_out_trace));
fprintf('Breakthrough (10%% of feed N2): t B10 = %.2f s\n', tB10);

```

```

fprintf('Breakthrough (50%% of feed N2): t_B50 = %.2f s\n', tB50);
if ~isnan(MTZ), fprintf('Estimated MTZ length at t_B50: %.3f m\n', MTZ); end

yN2_end = Y(end,1:Nx); T_end = Y(end,3*Nx+1:4*Nx);
prof_end = compute_pressure_velocity_x(pars, yN2_end, T_end);
Ctot_L = prof_end.Ctot(end);
yO2_out = 1 - yN2_end(end);
C_O2_out = yO2_out*Ctot_L; C_N2_out = (1-yO2_out)*Ctot_L;
purityC = 100*C_O2_out/(C_O2_out + C_N2_out);
fprintf('Outlet O2 concentration (x=L, final): %.2f mol/m^3; Purity: %.1f%%\n', C_O2_out, purityC);

function tB = localFirstCross(ytrace, tvec, thr)
    idx = find(ytrace >= thr, 1, 'first');
    if isempty(idx)
        tB = NaN;
    elseif idx == 1
        tB = tvec(1);
    else
        y0c = ytrace(idx-1); y1c = ytrace(idx);
        t0c = tvec(idx-1); t1c = tvec(idx);
        if y1c == y0c
            tB = t1c;
        else
            tB = t0c + (thr - y0c)*(t1c - t0c)/(y1c - y0c);
        end
    end
end

function dy = rhs_yN2_xT(~,y,par)
    Nx = par.Nx; dx = par.dx; eps = par.eps; rho_p = par.rho_p;
    kLDF = par.kLDF;
    qmaxN2 = par.qmaxN2; qmaxO2 = par.qmaxO2; bN2 = par.bN2; bO2 = par.bO2;

```

```

Dax = par.Dax; Cp_g_m = par.Cp_g_m; Cp_s = par.Cp_s; lambda_eff = par.lambda_eff;
dH_N2 = par.dH_N2; dH_O2 = par.dH_O2; yN2_in = par.yN2_in;

yN2 = y(1:Nx);
qN2 = y(Nx+1:2*Nx);
qO2 = y(2*Nx+1:3*Nx);
T = y(3*Nx+1:4*Nx);

prof = compute_pressure_velocity_x(par, yN2, T);
P = prof.P; U = prof.U; Ctot = prof.Ctot;

yN2 = max(1e-9, min(1-1e-9, yN2)); yO2 = 1 - yN2;

PN2 = yN2 .* P; PO2 = yO2 .* P;
den = 1 + bN2.*PN2 + bO2.*PO2;
qN2s = qmaxN2 * (bN2.*PN2) ./ den;
qO2s = qmaxO2 * (bO2.*PO2) ./ den;

dqN2dt = kLDF * (qN2s - qN2);
dqO2dt = kLDF * (qO2s - qO2);

d2y = zeros(Nx,1);
d2y(2:end-1) = (yN2(3:end) - 2*yN2(2:end-1) + yN2(1:end-2))/dx^2;
d2y(1) = (yN2(2) - yN2(1))/dx^2;
d2y(end) = (yN2(end-1) - yN2(end))/dx^2;

dy_dx = zeros(Nx,1);
dy_dx(1) = (yN2(1) - yN2_in)/dx;
dy_dx(2:end) = (yN2(2:end) - yN2(1:end-1))/dx;

dyN2dt = ( Dax.*Ctot .* d2y - U.*Ctot .* dy_dx - ((1-eps)/eps)*rho_p.*dqN2dt ) ./ (eps.*Ctot);

Ce = eps.*Ctot.*Cp_g_m + (1-eps).*rho_p.*Cp_s;

```

```

d2T = zeros(Nx,1);
d2T(2:end-1) = (T(3:end) - 2*T(2:end-1) + T(1:end-2))/dx^2;
d2T(1) = (T(2) - T(1))/dx^2;
d2T(end) = (T(end-1) - T(end))/dx^2;

dT_dx = zeros(Nx,1);
dT_dx(1) = 0;
dT_dx(2:end) = (T(2:end) - T(1:end-1))/dx;

Qads = (1-eps).*rho_p .* ( dH_N2.*dqN2dt + dH_O2.*dqO2dt );

dTdt = ( lambda_eff.*d2T + Qads - (U.*Ctot.*Cp_g_m).*dT_dx ) ./ Ce;

dy = [dyN2dt; dqN2dt; dqO2dt; dTdt];
end

function prof = compute_pressure_velocity_x(par, yN2, T)
    Nx = par.Nx; dx = par.dx; %ok<NASGU>
    R = par.R; eps = par.eps;
    dp = par.dp; mu = par.mu; P_in = par.P_in; u_in = par.u_in;

    yN2 = max(1e-9, min(1-1e-9, yN2)); yO2 = 1 - yN2;
    M_N2 = 0.0280134; M_O2 = 0.031998;
    Mmix = yN2*M_N2 + yO2*M_O2;

    Ctot_in = P_in/(R*T(1));
    Gm = u_in * Ctot_in;

    P = zeros(Nx,1);
    U = zeros(Nx,1);
    Ctt = zeros(Nx,1);
    P(1) = P_in;

```

```

for i = 1:Nx-1
    Ctt(i) = P(i)/(R*T(i));
    U(i) = Gm / Ctt(i);
    rho_g = Ctt(i) * Mmix(i);

    visc = 150*((1-eps)^2/eps^3) * mu * U(i) / (dp^2);
    inert = 1.75*((1-eps)/eps^3) * rho_g * U(i)^2 / dp;
    dPdx = -(visc + inert);

    P(i+1) = max(1e3, P(i) + dPdx*(par.dx));
end

Ctt(end) = P(end)/(R*T(end));
U(end) = Gm / Ctt(end);

prof = struct('P',P,'U',U,'Ctot',Ctt);
end

```

Graphical Representations Code

```

function Graphical_Representationsq
R = 8.314; T_in = 298.0; Pbar = 4.5; P_in = Pbar*1e5;
yH2_in = 0.79;

L = 0.55; eps = 0.37; rho_p = 1100.0; dp = 2e-3;
u_in = 0.09; Dax = 7.5e-5; kLDF = 0.11; mu = 1.8e-5;
lambda_eff = 0.15; Cp_g_m = 29.0; Cp_s = 900.0;
dH_H2 = 2.0e4; dH_O2 = 1.2e4;

qmaxH2 = 1.8; bH2bar = 0.07;
qmaxO2 = 0.45; bO2bar = 0.02;
bH2 = bH2bar/1e5; bO2 = bO2bar/1e5;

Nx = 80; x = linspace(0,L,Nx).'; dx = L/(Nx-1);
tend = 60.0;

yH2_0 = zeros(Nx,1); qH2_0 = zeros(Nx,1);
qO2_0 = zeros(Nx,1); T0 = T_in*ones(Nx,1);
y0 = [yH2_0; qH2_0; qO2_0; T0];

pars = struct('Nx',Nx,'dx',dx,'eps',eps,'rho_p',rho_p,'R',R, ...
'Dax',Dax,'kLDF',kLDF,'qmaxH2',qmaxH2,'qmaxO2',qmaxO2, ...
'bH2',bH2,'bO2',bO2,'yH2_in',yH2_in, ...
'dp',dp,'mu',mu,'Cp_g_m',Cp_g_m,'Cp_s',Cp_s, ...
'lambda_eff',lambda_eff,'dH_H2',dH_H2,'dH_O2',dH_O2, ...
'P_in',P_in,'u_in',u_in,'T_in',T_in);

opts = odeset('RelTol',1e-6,'AbsTol',1e-8,'MaxStep',0.25);
[tSol, Y] = ode15s(@(t,y) rhs_yH2_xf_fu(t,y,pars), [0 tend], y0, opts);

```

```

yH2_out_trace = Y(:,1:Nx,Nx);
yH2_out_trace = yH2_out_trace(:,end);
thr10 = 0.10*yH2_in; thr50 = 0.50*yH2_in;
tB10 = localFirstCross_fu(yH2_out_trace, tSol, thr10);
tB50 = localFirstCross_fu(yH2_out_trace, tSol, thr50);

snap1 = [0, 10, 20, 40, tend];
idx = arrayfun(@(ts) find(abs(tSol-ts)==min(abs(tSol-ts)),1,'first'), snap1);

yH2_end = Y(end,1:Nx).';
T_end = Y(end,3*Nx+1:4*Nx).';
prof_end = compute_pressure_velocity_x_fu(pars, yH2_end, T_end);
P_end = prof_end.P;
Ctot_end = prof_end.Ctot;

figure('Name','Temperature profile'); hold on; grid on;
for m = 1:numel(idx)
    t_snap = Y(idx(m), 3*Nx+1:4*Nx).';
    plot(x, t_snap, 'linewidth', 1.5);
end
xlabel('Axial position, x (m)');
ylabel('Temperature, T (K)');
legend(arrayfun(@(ts) sprintf('t = %.0f s', ts), snap1, 'UniformOutput', false), ...
'location','best');
title('Temperature profile along the bed');

figure('Name','Pressure drop'); grid on;
plot(x, P_end/1e5, 'linewidth', 1.8);
xlabel('Axial position, x (m)');
ylabel('Pressure, P (bar)');
dPbar = (P_end(1)-P_end(end))/1e5;
title(sprintf('Pressure profile at final time (\Delta P = %.2f bar)', dPbar));

figure('Name','Species profile'); hold on; grid on;
yH2_f = yH2_end;
yO2_f = 1 - yH2_f;
plot(x, yH2_f, 'linewidth', 1.8);
plot(x, yO2_f, 'linewidth', 1.8);
xlabel('Axial position, x (m)');
ylabel('Mole fraction, y');

```

```

legend('y_0.2(s)','y_0.2(s)','location','best');
title('Composition from inlet (x=0) to outlet (x=L) at final time');

figure('Name','breakthrough'); hold on; grid on;
plot(totl, y02_out_trace, 'linewidth', 1.0);
yl = ylim;
if ~isnan(t010)
    plot([t010 t010], yl, '-.', 'linewidth', 1.0);
end
if ~isnan(t050)
    plot([t050 t050], yl, '-.', 'linewidth', 1.0);
end
xlabel('Time, t (s)'); ylabel('Outlet y_0.2 (-)');
leg = {'y_0.2,out'(t)};
if ~isnan(t010), leg(end+1)=sprintf('t_010=%3.2f s', t010); end
if ~isnan(t050), leg(end+1)=sprintf('t_050=%3.2f s', t050); end
legend(leg,'location','best');
title('breakthrough behaviour at x = L (air in at x = 0)');
end

function dy = rho_y02_x_fv(par, y02, y, par)
    Re = par.Re; dx = par.dx; eps = par.eps; rho_p = par.rho_p;
    k1Df = par.k1Df;
    qmax02 = par.qmax02; qmax02 = par.qmax02; b02 = par.b02; b02 = par.b02;
    Dm = par.Dm; Cp_gm = par.Cp_gm; Cp_s = par.Cp_s; lambda_eff = par.lambda_eff;
    d1_02 = par.d1_02; d1_02 = par.d1_02; y02_in = par.y02_in;

    y02 = y(1:Nx);
    q02 = y(Nx+1:2*Nx);
    q02 = y(2*Nx+1:3*Nx);
    T = y(3*Nx+1:4*Nx);

    prof = compute_pressure_velocity_x_fv(par, y02, T);
    P = prof.P; U = prof.U; Ctot = prof.Ctot;

    y02 = max(1e-9, min(1-1e-9, y02)); y02 = 1 - y02;

    P02 = y02 .* P; P02 = y02 .* P;

```

```

den = 1 + b02.*y02 + b02.*q02;
q02s = qmax02 * (b02.*y02) ./ den;
q02s = qmax02 * (b02.*q02) ./ den;

dq02dt = k1Df * (q02s - q02);
dq02dt = k1Df * (q02s - q02);

dy = zeros(Nx,1);
dy(2:end-1) = (q02(2:end) - 2*q02(2:end-1) + q02(1:end-2))/dx^2;
dy(1) = -(y02(2) - y02(1))/dx^2;
dy(end) = (y02(end-1) - y02(end))/dx^2;

dy_dx = zeros(Nx,1);
dy_dx(1) = (y02(2) - y02(1))/dx;
dy_dx(2:end) = (y02(2:end) - y02(1:end-1))/dx;

dy02dt = ( Dm.*Ctot .* dy - U.*Ctot - ((1-eps)/eps)*rho_p.*dy02dt ) ./ (eps.*Ctot);

Ce = eps.*Ctot.*Cp_gm + (1-eps).*rho_p.*Cp_s;

dT = zeros(Nx,1);
dT(2:end-1) = (T(2:end) - 2*T(2:end-1) + T(1:end-2))/dx^2;
dT(1) = -(T(2) - T(1))/dx^2;
dT(end) = (T(end-1) - T(end))/dx^2;

dT_dx = zeros(Nx,1);
dT_dx(1) = T(2) - T(1)/dx;
dT_dx(2:end) = (T(2:end) - T(1:end-1))/dx;

Q02s = (1-eps).*rho_p * ( d1_02.*dq02dt + d1_02.*dq02dt );

dTdt = ( lambda_eff.*dT + Q02s - (U.*Ctot.*Cp_gm).*dT_dx ) ./ Ce;

dy = dy02dt; dq02dt; dq02dt; dTdt;

end

function prof = compute_pressure_velocity_x_fv(par, y02, T)
    Re = par.Re; dx = par.dx; k = par.k; eps = par.eps;
    dp = par.dp; dm = par.dm; P_in = par.P_in; u_in = par.u_in;

    y02 = max(1e-9, min(1-1e-9, y02)); y02 = 1 - y02;
    P_02 = 0.0200134; P_02 = 0.031908;

```

```

Ctot_in = P_in/(R*T(1));
Gm = u_in * Ctot_in;

P = zeros(Nx,1); U = zeros(Nx,1); Ctt = zeros(Nx,1);
P(1) = P_in;

for i = 1:Nx-1
    Ctt(i) = P(i)/(R*T(i));
    U(i) = Gm / Ctt(i);
    rho_g = Ctt(i) * (y02(i)*M_02 + (1-y02(i))*M_02);

    visc = 150*((1-eps)^2/eps^3) * mu * U(i) / (dp^2);
    inert = 1.72*((1-eps)/eps^3) * rho_g * U(i)^2 / dp;
    dPdx = -(visc + inert);

    P(i+1) = max(1e3, P(i) + dPdx*dx);
end
Ctt(end) = P(end)/(R*T(end));
U(end) = Gm / Ctt(end);

prof = struct('P',P,'U',U,'Ctot',Ctt);

function t0 = localFirstCross_fw(ytrace, tvec, thr)
    idx = find(ytrace >= thr, 1, 'first');
    if isempty(idx)
        t0 = NaN;
    elseif idx == 1
        t0 = tvec(1);
    else
        y0c = ytrace(idx-1); y1c = ytrace(idx);
        t0c = tvec(idx-1); t1c = tvec(idx);
        if y1c == y0c
            t0 = t1c;
        else
            t0 = t0c + (thr - y0c)*(t1c - t0c)/(y1c - y0c);
        end
    end
end
end

```

Derivations

Gas-phase species balance in mole-fraction form

$$\varepsilon \frac{\partial c_i}{\partial t} + \frac{\partial}{\partial x} \left(u c_i - D_{ax} \frac{\partial c_i}{\partial x} \right) = -(1 - \varepsilon) \rho_p \frac{\partial q_i}{\partial t}$$

$$\varepsilon \frac{\partial C_{\text{tot}}}{\partial t} + \frac{\partial}{\partial x} \left(u C_{\text{tot}} - D_{ax} \frac{\partial C_{\text{tot}}}{\partial x} \right) = -(1 - \varepsilon) \rho_p \sum_i \frac{\partial q_i}{\partial t}$$

$$\varepsilon C_{\text{tot}} \frac{\partial y_i}{\partial t} + u C_{\text{tot}} \frac{\partial y_i}{\partial x} = D_{ax} C_{\text{tot}} \frac{\partial^2 y_i}{\partial x^2} - \frac{1 - \varepsilon}{\varepsilon} \rho_p \frac{\partial q_i}{\partial t}$$

LDF intraparticle rate from spherical diffusion

$$\frac{\partial q}{\partial t} = D_p \left(\frac{1}{r^2} \frac{\partial}{\partial r} \left[r^2 \frac{\partial q}{\partial r} \right] \right), \quad q(R_p, t) = q^*(t), \quad q(r, 0) = q_0$$

$$\bar{q}(t) - q^* \approx \exp(-\beta D_p t / R_p^2)$$

$$\frac{\partial \bar{q}}{\partial t} = k_{\text{LDF}}(q^* - \bar{q}), \quad k_{\text{LDF}} \approx \beta D_p / R_p^2, \quad \beta \simeq 15$$

Extended Langmuir isotherm

$$\theta_i = b_i P_i \theta_v, \quad \theta_v = \frac{1}{1 + \sum_j b_j P_j}$$

$$q_i^* = \frac{q_{\text{max},i} b_i P_i}{1 + b_{N_2} P_{N_2} + b_{O_2} P_{O_2}}$$

Single-temperature energy balance; effective heat capacity

$$\varepsilon C_{\text{tot}} C_{p,g}^{(m)} \frac{\partial T}{\partial t} + u C_{\text{tot}} C_{p,g}^{(m)} \frac{\partial T}{\partial x} = \lambda_{\text{eff}} \frac{\partial^2 T}{\partial x^2} + \dot{q}_{gs}$$

$$(1 - \varepsilon) \rho_p C_{p,s} \frac{\partial T}{\partial t} = -\dot{q}_{gs} + (1 - \varepsilon) \rho_p \sum_i (-\Delta H_i) \frac{\partial q_i}{\partial t}$$

$$C_e = \varepsilon C_{\text{tot}} C_{p,g}^{(m)} + (1 - \varepsilon) \rho_p C_{p,s}$$

$$C_e \frac{\partial T}{\partial t} + u C_{\text{tot}} C_{p,g}^{(m)} \frac{\partial T}{\partial x} = \lambda_{\text{eff}} \frac{\partial^2 T}{\partial x^2} + (1 - \varepsilon) \rho_p \sum_i (-\Delta H_i) \frac{\partial q_i}{\partial t}$$

Ergun closure; continuity $\rightarrow$ superficial velocity

$$C_{O_2,\text{out}} = y_{O_2,\text{out}} C_{\text{tot},L}, \quad C_{N_2,\text{out}} = (1 - y_{O_2,\text{out}}) C_{\text{tot},L}$$

$$\text{Purity}[\%] = 100 \frac{C_{O_2,\text{out}}}{C_{O_2,\text{out}} + C_{N_2,\text{out}}}$$

Breakthrough times; MTZ length

$$t_{B10} : y_{N_2,\text{out}} = 0.10 y_{N_2}^{\text{in}}, \quad t_{B50} : y_{N_2,\text{out}} = 0.50 y_{N_2}^{\text{in}}$$

$$t_B = t_{k-1} + (\text{thr} - y_{k-1}) \frac{t_k - t_{k-1}}{y_k - y_{k-1}}$$

$$\tilde{y}(x) = y_{N_2}(x, t_{B50}) / y_{N_2}^{\text{in}}$$

$$x_{10} : \tilde{y} = 0.10, \quad x_{90} : \tilde{y} = 0.90$$

$$\text{MTZ} = \max(0, x_{90} - x_{10})$$

